INCH-POUND

MIL-M-28787/47B(SH)
AMENDMENT 2
19 November 1990

SUPERSEDING
AMENDMENT 1
29 DECEMBER 1982

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

BUFFER, THREE-STATE, DIGITAL, KEY CODE BBB

This amendment forms a part of Military Specification MIL-M-28787/47B(EC), dated 7 March 1980, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 6

* TABLE II: Delete and substitute new table II as printed on page 2 of this amendment.

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4.3: Delete and substitute the following:

"4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25 ± 5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation, frame	III	S	10	–	1	–	10	–	1	–	MΩ
A2	Isolation, unused contacts	III	X	10	–	1	–	10	–	1	–	MΩ
B1-B12	Maximum stress current check	IV	S	–	–	–	–	–	–	–	–	–
C1-C12	V_{OUT} (low-loaded)	V	S	0.01	1.0	0.01	1.0	0.01	1.0	0.01	1.0	V
D1-D24	Input breakdown current (data and disable)	VI	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
D25	Input breakdown current (mode control)	VI	S	0.0	3.0	0.0	3.0	0.0	3.0	0.0	3.0	mA
E1-E12	Output short circuit current	VII	S	-40	-120	-40	-120	-40	-120	-40	-120	mA
F1-F25	Input clamp diode voltage	VIII	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
G1-G12	Output clamp diode voltage	VIII	X	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
H1-H12	Output leakage current, (high level)	IX	X	-40	40	-40	40	-40	40	-40	40	uA
J1-J12	Output leakage current, (power off)	IX	X	0	40	0	40	0	40	0	40	uA
K1-K12	Output leakage current, (low level)	X	X	-40	40	-40	40	-40	40	-40	40	uA
L1-L24	Output voltage, low level, I_{OL} = 32 mA	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
L25-L36	Output voltage, low level, I_{OL} = 40 mA	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
M1-M24	Output voltage, high level, I_{OH} = -2 mA	XI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
M25-M36	Output voltage, high level, I_{OH} = -40 mA	XI	X	1.8	4.5	1.8	4.5	1.8	4.5	1.8	4.5	V

See footnotes at end of table.

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* 4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through GG shall be performed in any order with the following exceptions. Test conditions A, B, C, D, E, and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D, E, or F are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, and F where applicable, followed by the remaining tests in the sequence."

* 4.5.1, fifth line: Delete "E through EE" and substitute "G through T18, U1-U18, V through Z, and EE".

* 4.5.3: Delete and substitute the following:

"4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests G through T18, U1-U18, V through Z, and EE shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table II. Following the return of the module to 25 ± 5°C, electrical tests G through T18, U1-U18, V through Z, and EE shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table II."

6.3: Delete and substitute the following:

"6.3 Cross reference of classification. Military Specification MIL-M-28787/47 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833638. Part number M28787/47-1 supersedes part numbers 2833638 and M28787/47."

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FIGURE 6, NOTE 4, second line: Delete "minute" and substitute "second".

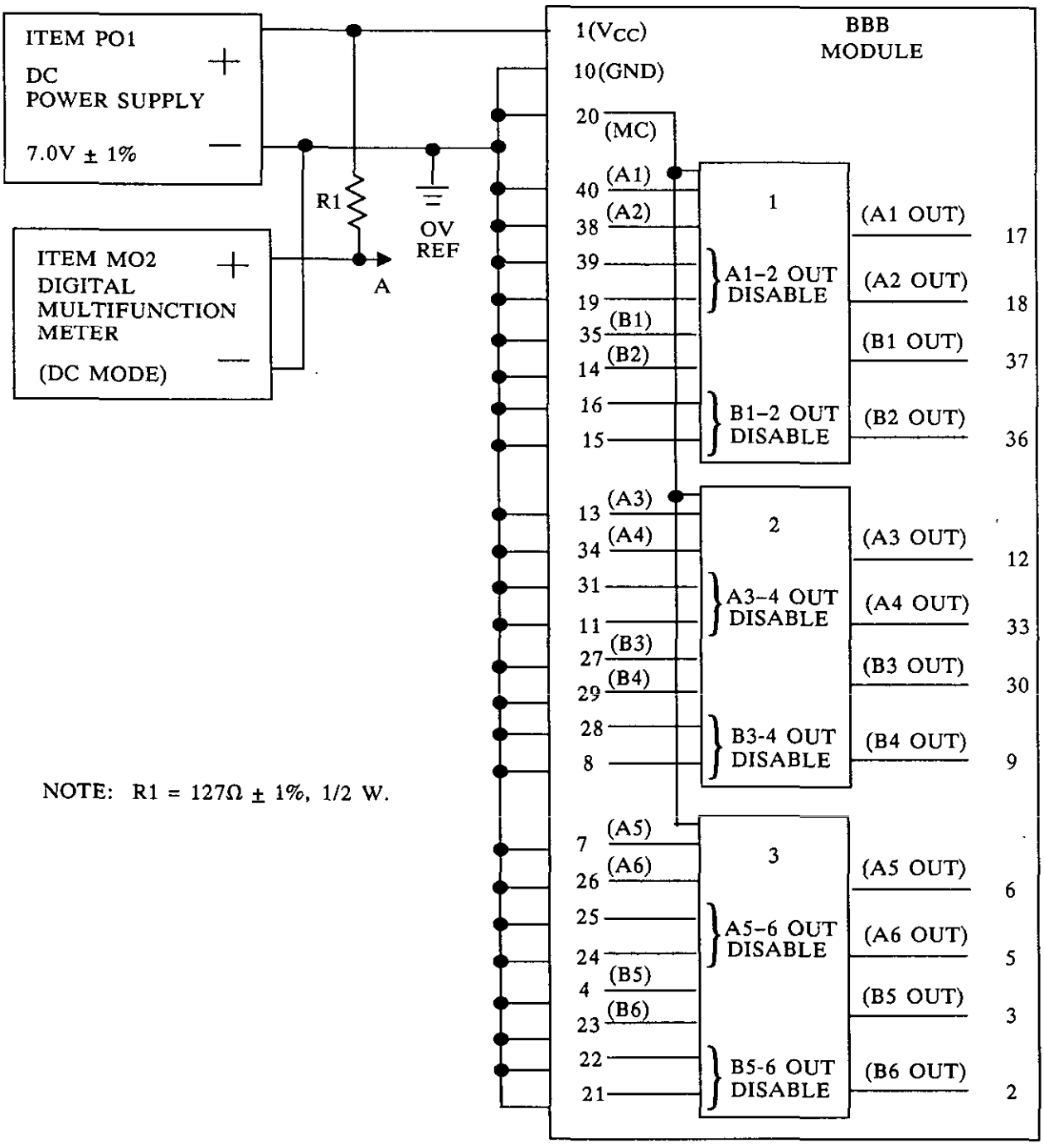
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FIGURE 7: Delete and substitute new figure 7 as printed on page 4 of this amendment.

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FIGURE 10: Delete and substitute new figure 10 as printed on page 5 of this amendment.

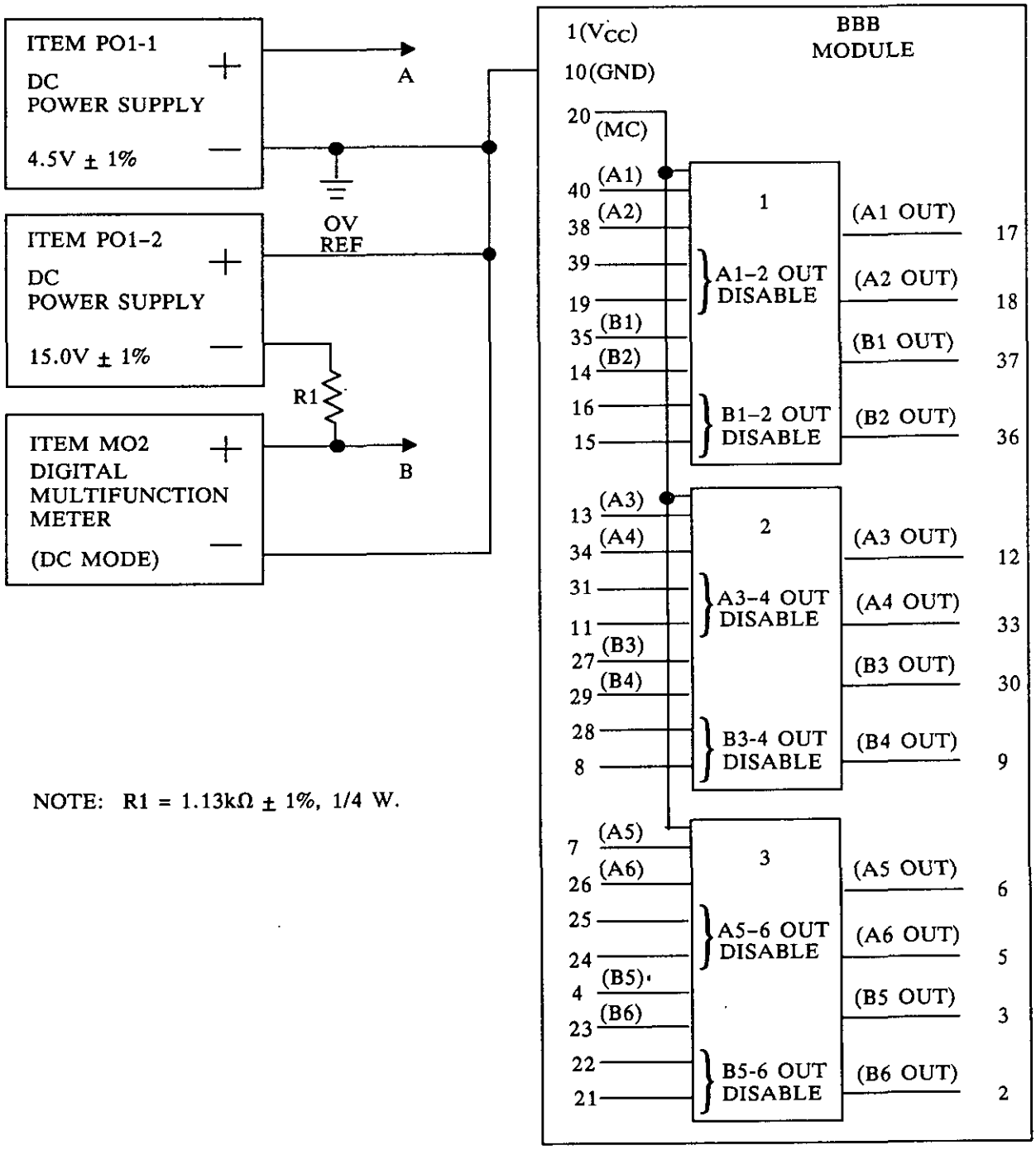
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NOTE: R1 = 127Ω ± 1%, 1/2 W.

FIGURE 7. V_{OUT} (low-loaded) test circuit for table V.

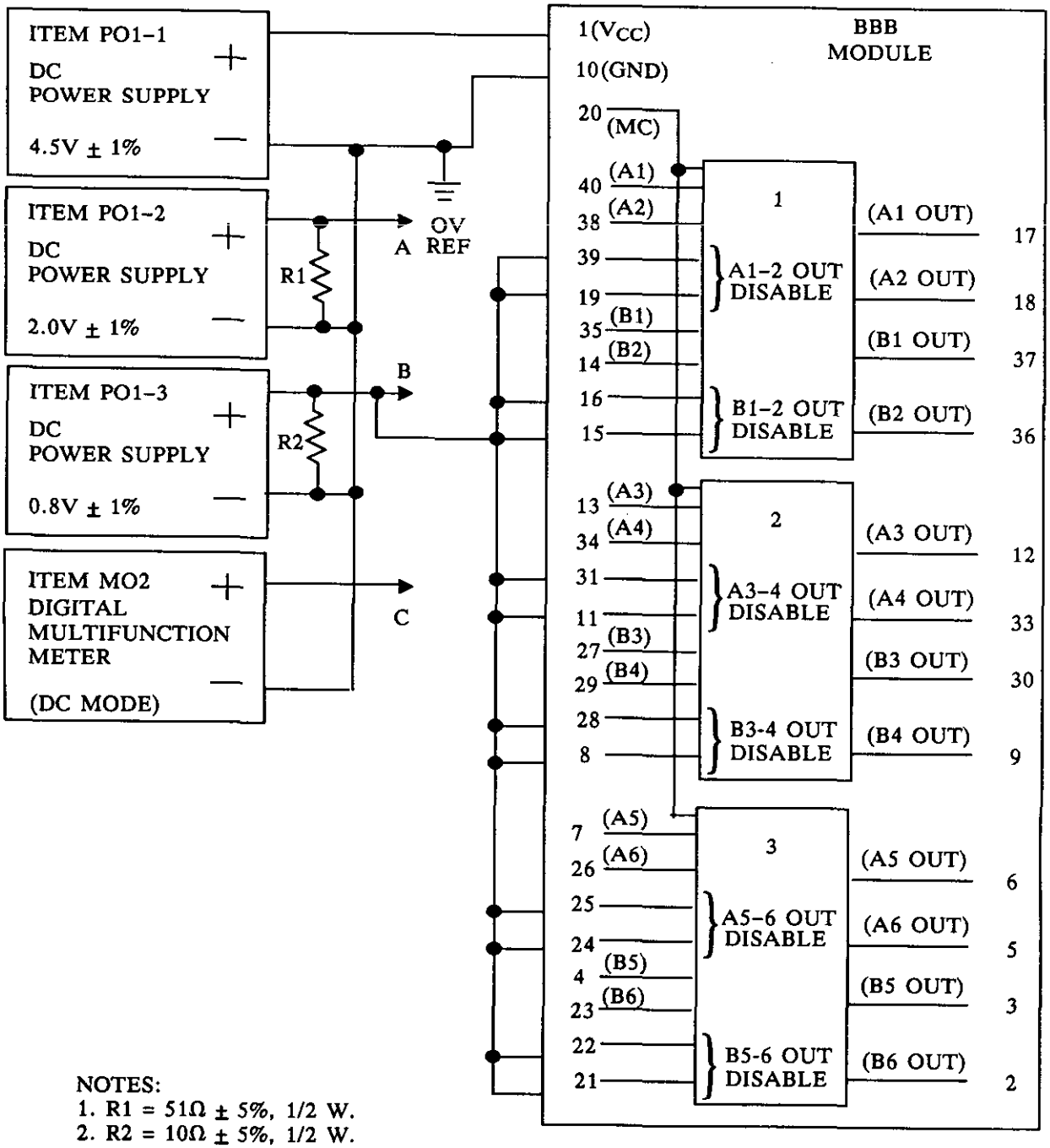
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NOTE: R1 = 1.13kΩ ± 1%, 1/4 W.

FIGURE 10. Input and output clamp diode voltage test circuit for table VIII.

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NOTES:
1. R1 = 51Ω ± 5%, 1/2 W.
2. R2 = 10Ω ± 5%, 1/2 W.
3. Load output under test with load as specified on figure 14.

FIGURE 13. Output voltage (low-level and high-level) test circuit for table XI.

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TABLE XI, Test condition M2, A to pin column: Delete "40" and substitute "38".

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FIGURE 13: Delete and substitute new figure 13 as printed on page 6 of this amendment.

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TABLE XVI, Test condition T14, A to pin column: Add "20" in column.

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TABLE XIX, Test condition AA5, C to pin column: Add "35" between 34 and 38.

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TABLE XIX, Test condition BB9, C to pin column: Add "35, 38, 40" after 34.

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy–SH

(Project 5963–N766)

Agent: NW

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